QP Code: 25900155

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC)- MG2DSCPHY100- MODERN PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

PART A

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

1. What is the difference between an inertial and non inertial frame of reference? [k]/[CO1]

Ans: An inertial frame of reference is one in which Newton's laws of motion holds. A frame of reference in which Newton's laws of motion does not hold is a non inertial frame of reference. **1 mark each for each point**

2. State 2 postulates of special relativity. [U]/[CO1]

Ans: The laws of physics are the same in all inertial frames of reference.

The speed of light in free space has the same value in all inertial frames of reference. **1 mark each for each postulate**

3**.** Explain the mass energy relation . [U]/[CO1]

Ans: Even when a body is at rest it possesses an energy known as rest mass energy. For a particle of rest mass m energy is given by $E = mc^2$, where c is the velocity of light in vacuum. **Eqn- 1 mark. Eqn & description 2 marks**

4. What are the features of blackbody? [K]/[CO2]

Ans: A body that absorbs all radiation incident upon it, regardless of frequency. A blackbody radiates more when it is hot than when it is cold, and the spectrum of a hot blackbody has its peak at a higher frequency than the peak in the spectrum of a cooler one. **Any 2 valid points related to blackbody- 2 marks**

Graph of Blackbody spectrum only 1 mark

5. What is the physical significance of the wavefunction? [K]/[CO2]

Ans: The wave function ψitself, has no direct physical significance. But $|\psi|^2$, the probability density is a measure of the probability of experimentally finding the body described by the wave function ψ at the point x, y, z, at the time t. **1 mark for each point**

6.Classify electromagnetic spectrum based on wavelength or frequency [U]/[CO2]

Ans:γ rays, X rays, Ultra Violet Rays, Visible light, Infrared Rays, Microwaves, Radio waves. **2 marks.**

Marks can be reduced proportionally if there is disorder.

7. In the photoelectric effect, what happens if the frequency of incident light is increased while keeping intensity constant? [U]/[CO2]

Ans: The higher the frequency of the light, the more energy the photoelectrons have. **2 marks**

8. What will be the ratio of de Broglie wavelength of a proton and an electron, if they have the same velocity? The ratio of their mass $m_p / m_e = 1836$ [A]/[CO2]

Ans: De Broglie wavelength $\lambda = \frac{h}{mv}$ **1 mark**

$\frac{\lambda_p}{\lambda_e} = \frac{m_e}{m_p} = \frac{1}{1836}$…….. **1 mark**

9. What is the effect of increasing the principal quantum number on an atom's energy levels? [K]/[CO3]

Ans: As the quantum number n increases, the corresponding energy E_n approaches closer to 0. The gap between energy levels becomes smaller. In the limit of $n = \infty , E_\infty = 0$ and the electron is no longer bound to the nucleus to form an atom. **2 mark**

Equation of energy only 1 mark

10.An astronomer finds a new absorption line with $\lambda = 164.1\ nm$ in the ultraviolet region of the Sun's continuous spectrum. He attributes the line to hydrogen's Lymann series. Is he right? Justify your answer.

[U]/[CO3]

Ans: The astronomer is wrong **1 mark**

The series limit of the Lyman series is 91 nm and 121 nm. The new absorption line is at 164.1 nm which is outside the limits of the Lyman series. **1 mark**

11. Define stimulated absorption in the context of laser physics. [U]/[CO4]

Ans: If the atom is initially in the lower state E0, it can be raised to E1 by absorbing a photon of energy $E_1 - E_0 = h\nu$. This process is called stimulated absorption. **2 marks**

Marks can be given if the pictorial representation is given

12.Write the steady state form of Schrodinger's equation in three dimensions [K]/[CO5]

$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2m}{\hbar^2}(E - U)\psi = 0$ **2 marks**

13. Give the physical interpretation of the wave function [U]/[CO5]

Ans: The quantity with which quantum mechanics is concerned is the wave function ψ of a body. While ψ itself has no physical interpretation, the square of its absolute magnitude $|\psi|^2$ evaluated at a particular place at a particular time is proportional to the probability of finding the body there at that time. **2 marks.**

14. A particle is confined in a 1D box of length L. Find the ratio of the energy of the first excited state to the energy of the ground state. [A]/[CO5]

Ans: For particle confined in a 1 Dimensional box, Energy $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$…….. **1 mark**

Ratio $\frac{E_1}{E_0} = 4$ **1 mark**

PART B

Short Essay Questions

Answer any 6 questions

Each question carries 5 marks

15. A train moving at 0.6 c has a proper length of 200 m. What is its observed length for a stationary observer? [U]/[CO1]

$L = L_0\sqrt{1 - \frac{v^2}{c^2}}$…… **2 mark**

$L_0 = 200\ m$

$v = 0.6c$

Substitution of values in the equation **1 mark**

$L = 160\ m$ …… **2 marks**

Reduce 1 mark for answer without unit.

16. A spaceship travels at 0.9c relative to Earth. If 1 hour passes on the spaceship, how much time passes on Earth? [A]/[CO1]

$T = \frac{T_0}{\sqrt{1-\frac{v^2}{c^2}}}$…… **2 mark**

$T_0 = 1\ Hr$

$v = 0.9c$

Substitution of values in the equation **1 mark**

$T = 2.29\ Hr$ …… **2 mark**

Reduce 1 mark for answer without unit.

17. Elaborate one example where Classical Physics failed and the new Quantum Physics solved the problem. [K]/[CO2]

Explanation of Black body radiation or Photoelectric effect **5 marks**

18. A microscope using photons is employed to locate an electron in an atom to within a distance of 0.2 Angstrom. What is the uncertainty in the momentum of the electron located in this way.?

[U]/[CO2]

$\Delta x \Delta p = \frac{\hbar}{2}$…… **2 marks**

$$\Delta x = 0.2 \times 10^{-10} m$$

Substitution of values in the equation **1 mark**

$\Delta p = 1.66 \times 10^{-23}$kg.m/s……. **2 marks**

Reduce 1 mark for answer without unit.

19. A metal has a work function of 3 eV. What is the longest wavelength of light that can cause photoelectric emission from this metal?

[A]/[CO2]

Work function $\phi = \frac{hc}{\lambda}$……. **2 mark**

Longest wavelength $\lambda = \frac{hc}{\phi}$

Substitution of values in the equation **1 mark**

$\lambda = 414\ nm$ …….. **2 marks**

Reduce 1 mark for answer without unit.

20. Discuss the process of energy absorption and emission in an atom. [K]/[CO3]

Ans: There are two main ways in which an atom can be excited to an energy above its ground state and thereby become able to radiate.

One of these ways is by a collision with another particle in which part of their joint kinetic energy is absorbed by the atom.

Such an excited atom will return to its ground state in an average of $10^{-8} s$ by emitting one or more photons.

Another excitation mechanism is involved when an atom absorbs a photon of light whose energy is just the right amount to raise the atom to a higher energy level.

When white light, which contains all wavelengths, is passed through hydrogen gas, photons of those wavelengths that correspond to transitions between energy levels are absorbed. The resulting excited hydrogen atoms reradiate their excitation energy almost at once, but these photons come off in random directions with only a few in the same direction as the original beam of white light. **Any one method-2.5, both 5 marks**

Just mention 2 methods- 2 marks. Pictorial representation only- 2 marks

21. Describe the process of population inversion and its role in the amplification of light in lasers.What methods are used to achieve it? [U]/[CO4]

Ans: The term population inversion describes an assembly of atoms in which the majority are in energy levels above the ground state. **2 marks.**

If we can arrange this and then shine light of frequency on the assembly, there will be more stimulated emissions from atoms in the metastable state than stimulated absorptions by atoms in the ground state. The result will be an amplification of the original light. This is the concept that underlies the operation of the laser. **1 mark**

Any two methods to attain population inversion (pumping method) **2marks**

22. Given the wavefunction for a free particle as $\psi = Ae^{(-\frac{i}{\hbar})(Et - px)}$. Obtain the time dependent form of Schrodinger equation.

[U]/[CO5]

Given $\psi = Ae^{(-\frac{i}{\hbar})(Et - px)}$

$$\frac{\partial^2 \psi}{\partial x^2} = Ae^{(-\frac{i}{\hbar})(Et - px)} \left(\frac{-p^2}{\hbar^2}\right)$$

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} = Ae^{(-\frac{i}{\hbar})(Et - px)} \left(\frac{p^2}{2m}\right) \text{.....(1)}$$

$$\frac{\partial \psi}{\partial t} = Ae^{(-\frac{i}{\hbar})(Et - px)} \left(\frac{-iE}{\hbar}\right)$$

$$i\hbar\frac{\partial \psi}{\partial t} = Ae^{(-\frac{i}{\hbar})(Et - px)} (E) \text{.....(2)}$$

Given the particle is a free particle. So $E = \frac{p^2}{2m}$

So from 1 and 2 we get $-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} = i\hbar\frac{\partial \psi}{\partial t}$

This is the time dependent form of Schrodinger equation for a free particle

correct derivation- 5 marks. Reduce marks for partially correct answers. Final Equation only- 2 marks